ARTIFICIAL INTELLIGENCE FOR NETWORK TRAFFIC MANAGEMENT

CSA0716-COMPUTER NETWORKS

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INTRODUCTION

Artificial Intelligence for Network Traffic Management

- AI improves the efficiency of modern computer networks.
- It monitors, analyses, and controls network traffic using machine learning algorithms.
- Widely used in data centres, automatically.
- AI helps reduce congestion and improves data transmission speed.
- It supports real-time decision-making loud computing, and 5G networks.

PROBLEM STATEMENT

- Challenges in Traditional Network Traffic Management
- Increasing network traffic causes congestion.
- Manual traffic management is slow and inefficient.
- Difficult to predict sudden traffic spikes.
- High latency and packet loss reduce network performance.
- Traditional methods cannot adapt quickly to changing network conditions.

Need for AI in Network Traffic Management

Why AI is Needed?

Rapid growth of internet users and connected devices.

Increasing demand for high-speed and reliable communication.

Dynamic traffic patterns require adaptive management.

Manual monitoring is time-consuming and prone to errors.

AI enables proactive traffic control and congestion prevention.

AI-BASED SOLUTION

How AI Manages Network Traffic

- Collects real-time traffic data from network devices.
- Uses Machine Learning to predict traffic patterns.
- Detects congestion before it occurs.
- Automatically reroutes traffic through optimal paths.
- Continuously learns and improves network performance.

APPLICATIONS AND BENEFITS

Applications:

Cloud Computing

5G and Wireless Networks

Internet Service Providers (ISPs)

Benefits:

Reduces network congestion

Improves bandwidth utilization

Lowers latency

Conclusion

- AI enables intelligent and automated network traffic management.
- It enhances network reliability, speed, and security.
- AI reduces human intervention while improving efficiency.

Thank You